Flexible Bayesian Models for Statistical Demography

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Slides: herbsusmann.com/ucd2026

Research programme

1. *Bayesian modelling:* flexible hierarchical models for demographic estimation and forecasting.

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2. *Causal inference:* semiparametric, machine-learning-based methods for observational data.

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Unifying agenda: Bayesian causal inference for statistical demography.

Research pillar: Bayesian modelling

Demographic and health indicators track population changes over time.

Example: modern contraceptive use rate

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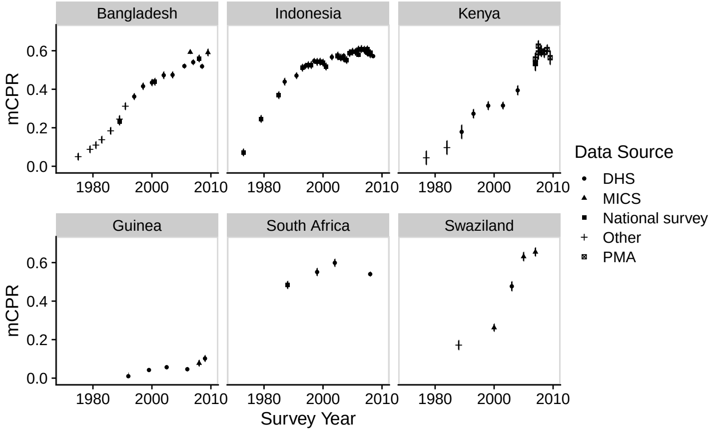
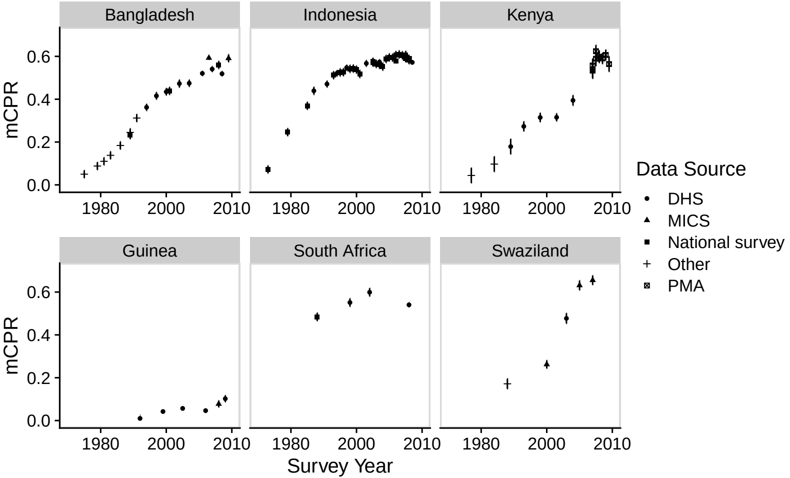
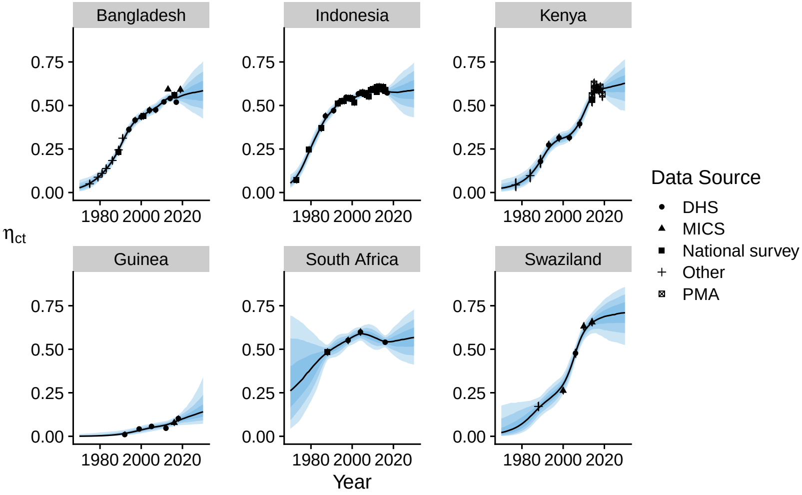


(Susmann and Alkema 2025 JRSS-C)

The challenge: from noisy, sparse data from multiple sources...



...produce probabilistic estimates and projections



Contribution: general modelling framework for demographic and health indicators

- **Challenge:** many statistical models have been created to provide estimates and projections, but...
  - comparing models can be difficult.
  - building a new model requires starting from scratch.

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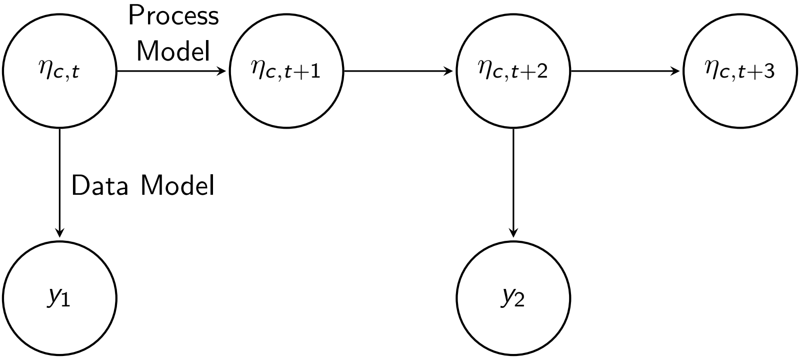
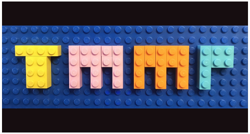
- **Challenge:** many statistical models have been created to provide estimates and projections, but...
  - comparing models can be difficult.
  - building a new model requires starting from scratch.
- **Contribution:** a modelling framework that decomposes models into interchangeable components, based on a common notation, that can be mixed and matched. (Susmann, Alexander, and Alkema, 2022 Int. Stat. Rev)

Temporal Models for Multiple Populations





Process Model

The **Process Model** defines how the true value of the indicator is expected to evolve over time.

$$g_1(\eta_{c,t}) = \underbrace{g_2(X_{c,t}, \beta_c)}_{\text{covariate}} + \underbrace{g_3(t, \eta_{c,s<t}, \alpha_c)}_{\text{systematic}} + \underbrace{\epsilon_{c,t}}_{\text{smoothing}}$$

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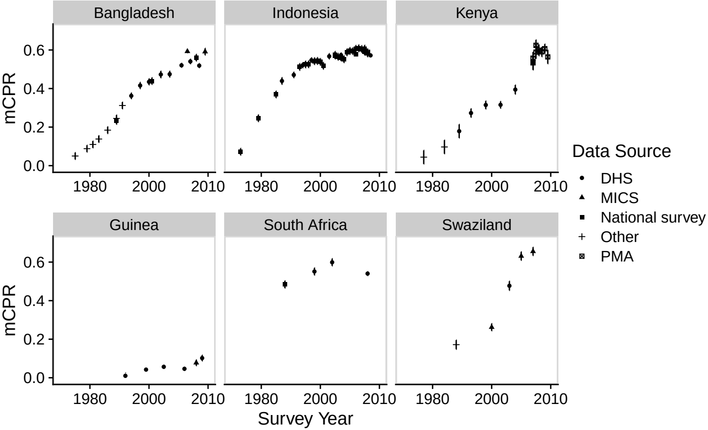
- **Systematic** $\rightarrow$ captures long-term expected trends.
- **Smoothing** $\rightarrow$ allows data-driven deviations from the other components, while still enforcing smoothness.

Contribution: modeling demographic transitions with B-splines

- **Challenge:** modeling indicators that follow *transitions* between stable states.



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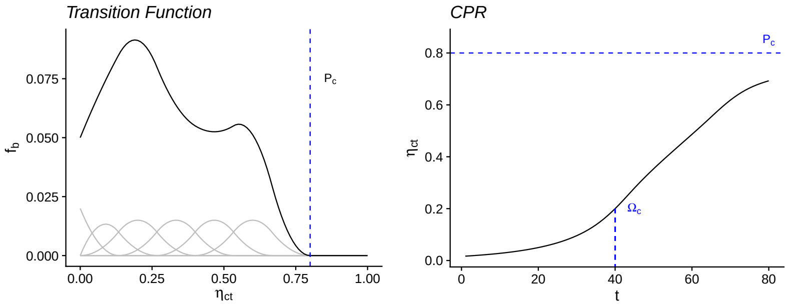
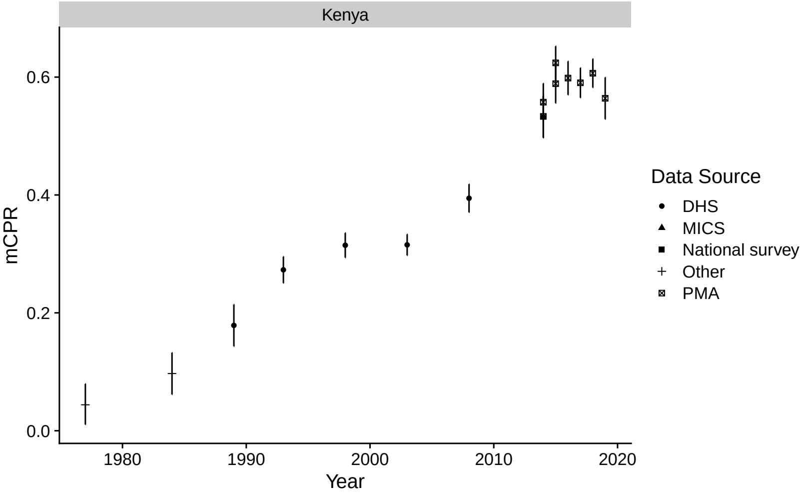
- **Contribution:** a general, flexible model applicable to a variety of indicators, without making strong functional form assumptions. (Susmann and Alkema, 2025 JRSS-C)
- **Key idea:** model the rate of change as a flexible B-spline function of the current level:

$$f_b(\eta_{c,t}, P_c, \boldsymbol{\beta}_c) = \sum_{j=1}^{J} \underbrace{h_j(\beta_{c,j})}_{\text{coefficient}} \cdot \underbrace{B_j(\eta_{c,t}/P_c)}_{\text{basis function}}.$$

B-spline transition functions capture a large class of smooth transitions

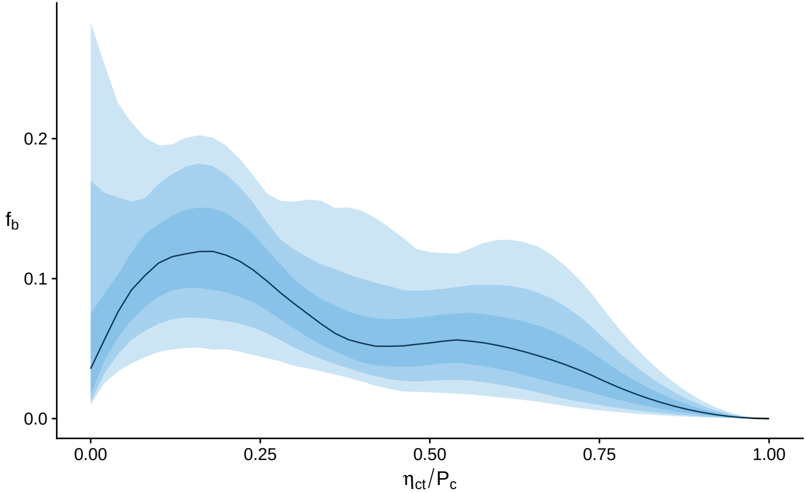


Kenya: raw data



$$\mathrm{logit}(\eta_{c,t}) = \underbrace{g_3(t, \eta_{c,s<t}, \alpha_c)}_{\text{systematic}} + \underbrace{\epsilon_{c,t}}_{\text{smoothing}}$$

Kenya: transition function



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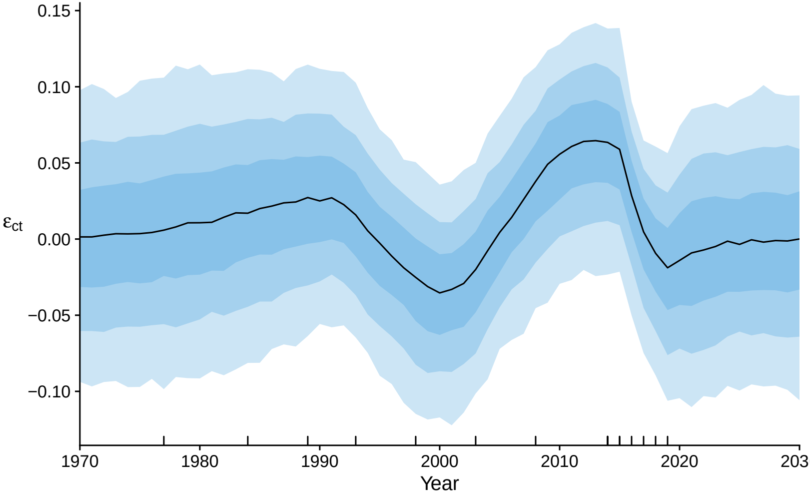
Kenya: smoothing component



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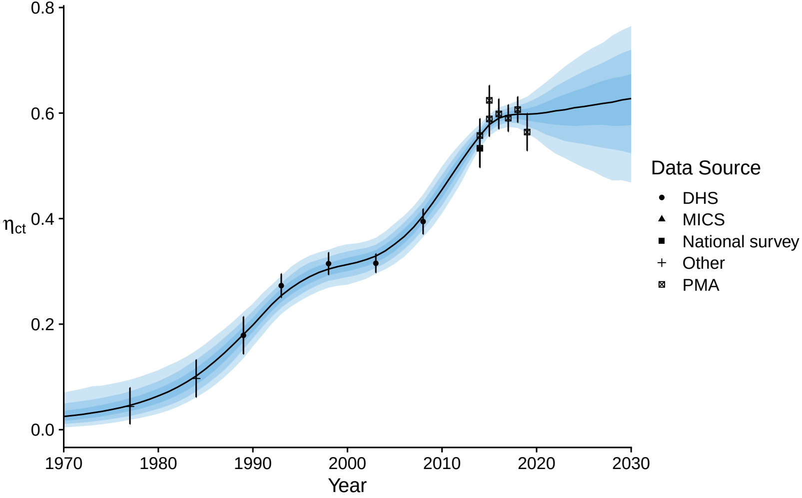
Kenya: Probabilistic estimates and projections



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Impact

- Presented at UN Population Division seminar (2025).
- B-spline transition model integrated in the *Family Planning Estimation Tool*: Used by countries to produce estimates of family planning indicators such as mCPR.
- Results used within countries and in the FP2030 global initiative.

FPET training by Avenir Health, Kenya 2024

Contribution: modeling refugee and asylum seeker populations with Interrupted Logistic Models

- Migration plays a key role in population projections:

  $$\text{Population}_{t+1} = \text{Population}_t + \text{Births}_t - \text{Deaths}_t + \text{Migration}_t$$

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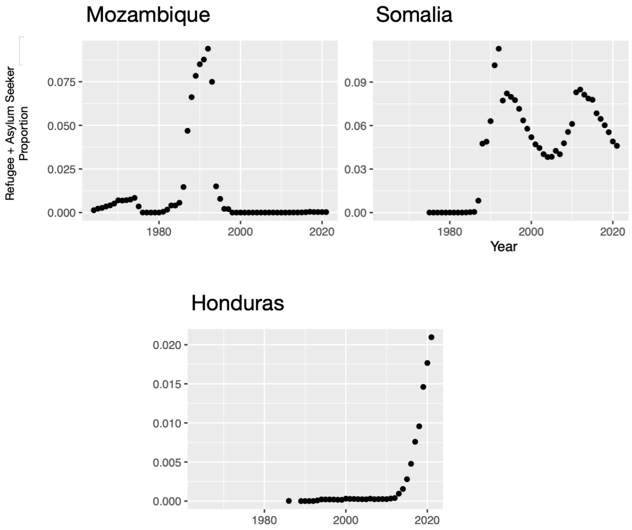
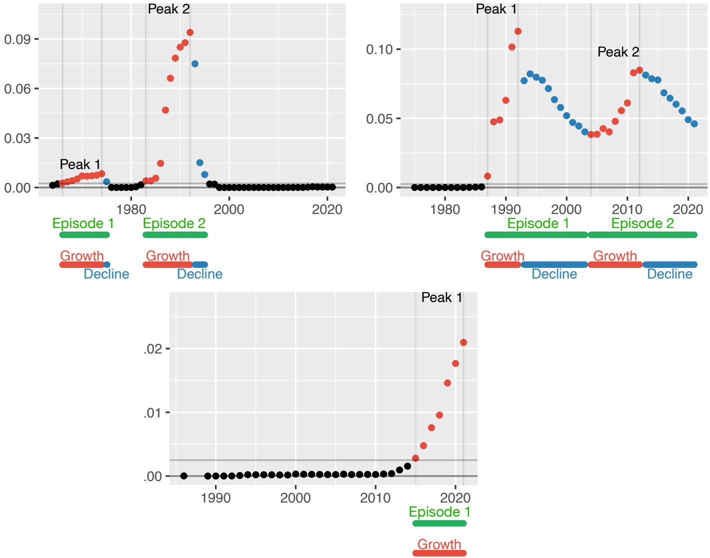
- **Challenge:** projecting refugee and asylum seeker populations is difficult due to highly contingent underlying processes.
- **Contribution:** a novel Bayesian model for refugee and asylum seeker populations. (Susmann and Raftery, 2025 Demography)

Refugee and asylum seeker population data

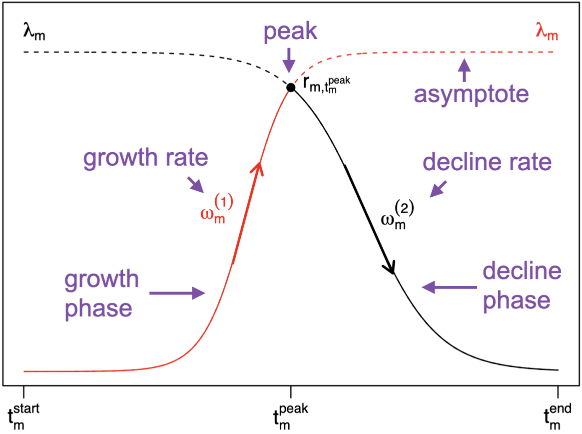


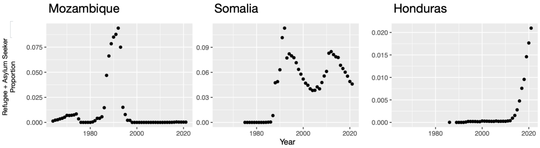
(Susmann and Raftery 2025 Demography; Data: United Nations High Commissioner for Refugees)

Segmentation into displacement episodes

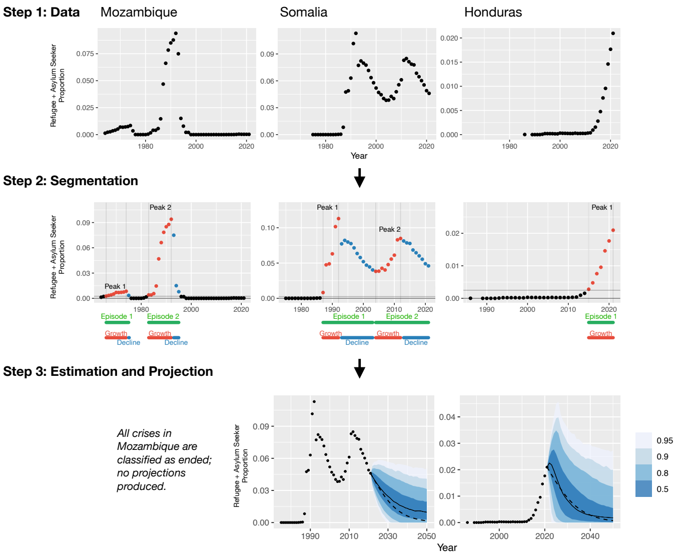


Interrupted Logistic Model: Structure





Integrated modeling pipeline



(Susmann and Raftery 2025 Demography)

Research pillar: Causal inference

Core theme: robust causal inference for complex observational data.

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1. **High dimensional exposures** $\rightarrow$ Nonparametric shrinkage estimators
    - Asymptotically efficient estimation of high-dimensional causal parameters.
    - Susmann et al. 2026, Annals of Applied Statistics

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2. **Complex longitudinal exposures** → Generalised Longitudinal Modified Treatment Policies
   - Extending the Average Treatment Effect on the Treated to time-varying, multi-valued, and continuous exposures.
   - Susmann et al. 2024, under review, Biometrics

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3. **Overlap violations** → Non-overlap bounds
   - Informative partial identification bounds when standard estimators fail.
   - Susmann, McClean, and Díaz 2025, preprint

Contribution: partial identification bounds for causal effects

Overlap assumption: every member of the population must have a positive probability of exposure.

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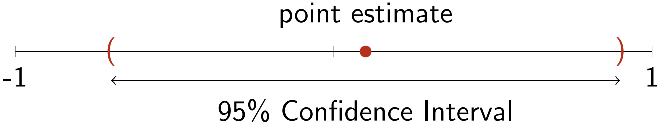
- ψ_O: estimated via doubly-robust methods; $\sqrt{n}$-consistent and semiparametrically efficient under weak conditions.
- ψ_{NO}: can be bounded under minimal assumptions.

Our proposal: estimate *bounds* on the treatment effect

- Traditional approach: estimate **point estimate** and **95% confidence interval**:



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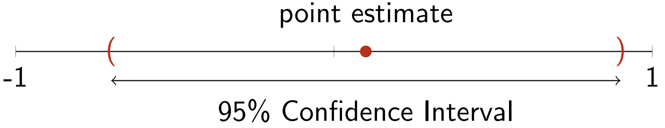
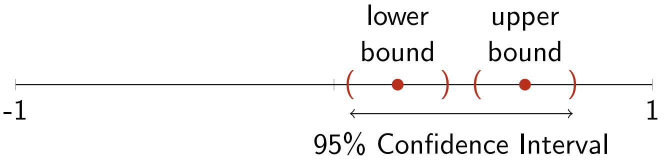


- Our approach: estimate **lower** and **upper bounds** and combine to form a **95% confidence interval**:



$$s_l(x,c,\gamma) = \begin{cases} 1, & x \leq c-\gamma, \\ 0, & x \geq c, \\ 1 - \exp\left[1 + \frac{1}{\{(x-c)/\gamma\}^2 - 1}\right], & \text{otherwise, and} \end{cases},$$

$$s_g(x,c,\gamma) = \begin{cases} 1, & x \geq c+\gamma, \\ 0, & x \leq c, \\ 1 - \exp\left[1 + \frac{1}{\{(x-c)/\gamma\}^2 - 1}\right], & \text{otherwise.} \end{cases}$$

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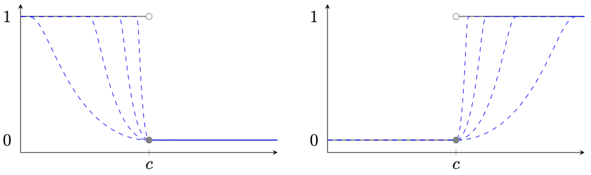
(A) smooth approximations of $\mathbb{I}(x < c)$ (B) smooth approximations of $\mathbb{I}(x > c)$



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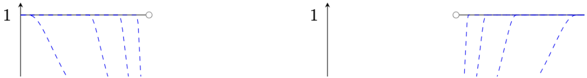

Theorem 1 (Efficient influence functions). *Under the setup of Proposition 2, suppose the maps $x \mapsto s_l(x, c, \gamma)$ and $x \mapsto s_g(x, c, \gamma)$ are twice-differentiable with bounded first and second derivatives. Let $\dot{s}$ denote the first derivative. Then, the parameters $\psi_s(c, \gamma)$, $L_s(c, \gamma)$, and $U_s(c, \gamma)$ admit von-Mises expansions of the form (6), with, respectively, the following uncentered efficient influence functions:*

$$
\begin{aligned}
\varphi_\psi(Z, \eta, c, \gamma) &= \mu_1 s_g(\pi, c, \gamma) - \mu_0 s_l(\pi, 1 - c, \gamma) \\
&\quad + \frac{A}{\pi} s_g(\pi, c, \gamma)\,(Y - \mu_1) - \frac{1 - A}{1 - \pi} s_l(\pi, 1 - c, \gamma)\,(Y - \mu_0) \\
&\quad + \{\mu_1 \dot{s}_g(\pi, c, \gamma) - \mu_0 \dot{s}_l(\pi, 1 - c, \gamma)\}\,(A - \pi)\,, \\
\varphi_L(Z, \eta, c, \gamma) &= \varphi_{\psi_s}(Z, \eta, c, \gamma) - 1 + s_l(\pi, 1 - c, \gamma) + \dot{s}_l(\pi, 1 - c, \gamma)\,(A - \pi)\,, \\
\varphi_U(Z, \eta, c, \gamma) &= \varphi_{\psi_s}(Z, \eta, c, \gamma) + 1 - s_g(\pi, c, \gamma) - \dot{s}_g(\pi, c, \gamma)\,(A - \pi)\,.
\end{aligned}
$$

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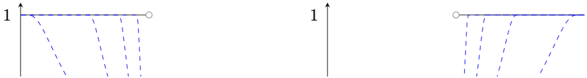

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Algorithm 2. *Given K threshold and smooth parameter combinations $\{(c_k,\gamma_k)\}_{k=1}^K$:*

1. *For each $k \in [K]$, construct efficient influence function estimates as in the prior section: $\{\varphi_{L,k}(Z_i,\widehat{\eta}), \varphi_{U,k}(Z,\widehat{\eta})\}_{i=1}^n$. Also construct point estimates and standard error estimators for the upper and lower bounds: $\left\{\widehat{L}_k, \widehat{U}_k\right\}$ and $\{\widehat{\sigma}_{L,k}, \widehat{\sigma}_{U,k}\}$.*

2. *For B bootstrap samples, draw i.i.d. multipliers $\{\xi_i^{(b)}\}_{i=1}^n$ with $\mathbb{E}(\xi)=0$, $\mathbb{E}(\xi^2)=1$, and form the studentized residuals for each index k:*

$$T^{(b)} \quad \frac{1}{\quad} \sum^{n} \xi^{(b)} \left\{ \varphi_{L,k}(Z_i,\widehat{\eta}) - \widehat{L}_k \right\} \qquad T^{(b)} \quad \frac{1}{\quad} \sum^{n} \xi^{(b)} \left\{ \varphi_{U,k}(Z_i,\widehat{\eta}) - \widehat{U}_k \right\}$$

arXiv > stat > arXiv:2509.20206

Statistics > Methodology

[Submitted on 24 Sep 2025]

Non-overlap Average Treatment Effect Bounds

Herbert P. Susmann, Alec McClean, Iván Díaz

The average treatment effect (ATE), the mean difference in potential outcomes under treatment and control, is a canonical causal effect. Overlap, which says that all subjects have non-zero probability of either treatment status, is necessary to identify and estimate the ATE. When overlap fails, the standard solution is to change the estimand, and target a trimmed effect in a subpopulation satisfying overlap; however, this no longer addresses the original goal of estimating the ATE. When the outcome is bounded, we demonstrate that this compromise is unnecessary. We derive non-overlap bounds: partial identification bounds on the ATE that do not require overlap. They are the sum of a trimmed effect within the overlap subpopulation and worst-case bounds on the ATE in the non-overlap subpopulation. Non-overlap bounds have width proportional to the size of the non-overlap subpopulation, making them informative when overlap violations are limited -- a common scenario in practice. Since the bounds are non-smooth functionals, we derive smooth approximations of them that contain the ATE but can be estimated using debiased estimators leveraging semiparametric efficiency theory. Specifically, we propose a Targeted Minimum Loss-Based estimator that is $\sqrt{n}$-consistent and asymptotically normal under nonparametric assumptions on the propensity score and outcome regression. We then show how to obtain a uniformly valid confidence set across all trimming and smoothing parameters with the multiplier bootstrap. This allows researchers to consider many parameters, choose the tightest confidence interval, and still attain valid coverage. We demonstrate via simulations that non-overlap bound estima ... TEs with h... wer than traditional doubly-robust point estimators. We illustrate ... e A... heterization on mortality.

A

n:

or

2. *For B bootstrap sample ... e ...* $\mathbb{E}(\xi) = 0$, $\mathbb{E}(\xi^2) = 1$, *and form the studentized residual ...*

$$T^{(b)} \quad \frac{1}{\ } \sum^{n} \xi^{(b)} \quad T^{(\ldots)} \quad \frac{1}{\ } \sum^{n} \xi^{(b)} \left\{ \varphi_{U,k}(Z_i, \widehat{\eta}) - \widehat{U}_k \right\}$$

Research vision: Bayesian modelling meets causal inference for statistical demography

- **Bayesian modelling:** How have populations changed, and how will they change?
- **Causal inference:** How *would* trends change under counterfactual interventions or policies?

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Summary

Two research pillars:

1. **Bayesian modelling** for demographic estimation and projection
   - Modular framework (Int. Stat. Review), flexible B-spline transition models (JRSS-C), refugee modelling (Demography).
   - Methods adopted in Family Planning Estimation Tool; presented at UN Population Division.
2. **Causal inference** for realistic observational data
   - Non-overlap bounds (preprint), generalised longitudinal effects (Biometrics, rev.), penalised estimation (AOAS).

Research vision: unify both pillars to advance Bayesian causal inference methodology for statistical demography.

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Transition Models

- A model class for indicators that follow a transition.
- *Transition Models* have a process model given by

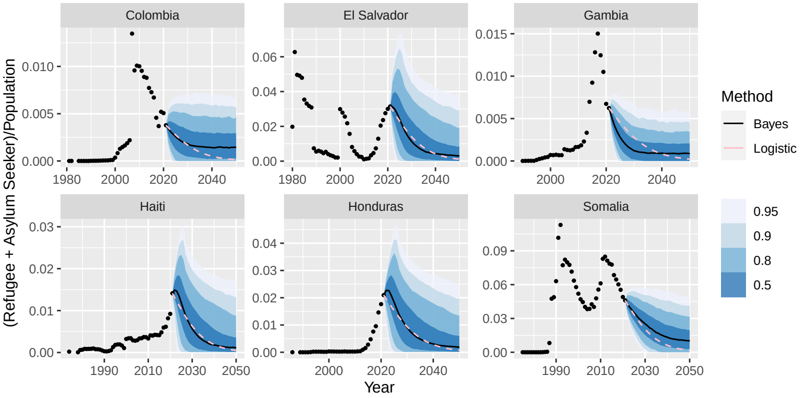
$$g_1(\eta_{c,t}) = \underbrace{g_3(t, \boldsymbol{\eta}_{c,s\leq t}, \boldsymbol{\alpha}_c)}_{\text{systematic}} + \underbrace{\epsilon_{c,t}}_{\text{smoothing}}.$$

- The systematic component has the following form:

$$g_3(t, \boldsymbol{\eta}_{c,s\neq t}, \boldsymbol{\alpha}_c) = \begin{cases} \Omega_c, & t = t_c^*, \\ g_1(\eta_{c,t-1}) + f(\eta_{c,t-1}, P_c, \boldsymbol{\beta}_c), & t > t_c^*, \\ g_1(\eta_{c,t+1}) - f(\eta_{c,t+1}, P_c, \boldsymbol{\beta}_c), & t < t_c^*, \end{cases}$$

  where $\boldsymbol{\alpha}_c = \{\Omega_c, P_c, \boldsymbol{\beta}_c\}$.
- The function f is called the *transition function*.

Projections for six illustrative countries



How the bounds are constructed

- Divide the population into overlap and non-overlap subpopulations.
- Set an *overlap threshold* c.
  - If $\pi \in [c, 1-c]$ then unit is in the *overlap* subpopulation.
  - If $\pi \notin [c, 1-c]$ then unit is in the *non-overlap* subpopulation.
- Assume that the outcome is *bounded*: $0 \leq Y \leq 1$.
- The ATE is bounded by:

$$
\begin{aligned}
\psi &= (Y^1 - Y^0) \\
&= (Y^1 - Y^0 \mid \pi \in [c, 1-c])\mathbb{P}(\pi \in [c, 1-c]) \\
&\quad + (Y^1 - Y^0 \mid \pi \notin [c, 1-c])\mathbb{P}(\pi \notin [c, 1-c]) \\
&\in (Y^1 - Y^0 \mid \pi \in [c, 1-c])\mathbb{P}(\pi \in [c, 1-c]) \\
&\quad \pm \mathbb{P}(\pi \notin [c, 1-c]).
\end{aligned}
$$

Non-overlap bounds, formally

Let $\pi(X) = \mathbb{P}(A = 1 \mid X)$, and fix $c \in \left[0, \frac{1}{2}\right]$ denote a propensity score trimming threshold. Under consistency, no unmeasured confounding, and outcome boundedness,

$$\left(Y^1 - Y^0\right) \in \Big[L(c), U(c)\Big].$$

where

$$\begin{aligned} L(c) &= \psi(c) - \mathbb{P}((X) \geq 1 - c), \\ U(c) &= \psi(c) + \mathbb{P}((X) \leq c), \text{ and} \\ \psi(c) &= \left\{\mu_1(X)\mathbb{I}(\pi(X) > c) - \mu_0(X)\mathbb{I}(\pi(X) < 1 - c)\right\}. \end{aligned}$$

Practical benefits: higher precision, higher power

	Mean Width		Power	
N	Doubly-robust	Bounds	Doubly-robust	Bounds
(A) Simulations conditional on non-overlap				
100	1.91	**0.46**	0.02	**0.49**
250	1.83	**0.29**	0.07	**0.88**
500	1.65	**0.22**	0.16	**1.00**
1000	1.35	**0.17**	0.31	**1.00**